

Measurement of $H \rightarrow WW^*$ in the $q\bar{q} \ell \nu q\bar{q}'$ final state at FCC-ee

Jinbo Zhang¹, Jan Eysermans² and Sara Aumiller¹

¹Department of Physics & Astronomy, University of California, Irvine, CA, USA

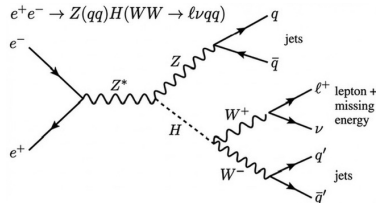
²Department of Physics, Massachusetts Institute of Technology, Cambridge, MA, USA

Signal

- $e^+e^- \rightarrow ZH, H \rightarrow WW^*$
- $Z \rightarrow q\bar{q}, W \rightarrow \ell\nu, W^* \rightarrow q\bar{q}'$
- Final state: 4 jets + 1 lepton + MET
- $\Sigma = 0.029 \text{ pb combined, } \sim 3.1 \times 10^5 \text{ events}$

Background

- Diboson: WW dominant, $\sigma = 16.4 \text{ pb}$; ZZ, $\sigma = 1.36 \text{ pb}$
- Two-fermion: $q\bar{q}, \sigma \sim 55 \text{ pb}$; $\tau\tau, \sigma = 4.67 \text{ pb}$
- Irreducible: ZH, with $H \rightarrow b\bar{b}, \tau\tau, gg, c\bar{c}, ZZ^*$



Process	Generator	σ [pb]	N_{gen}	Sample tag(s)
Signal: $e^+e^- \rightarrow ZH, H \rightarrow WW^*$				
$Z(q\bar{q})H(WW^*)$	Whizard	0.01140	1,100,000	uzp6_ee_qqH_WW_ecm240
$Z \rightarrow s\bar{s}/d\bar{d}$	Whizard	0.00635	1,000,000	uzp6_ee_bbH_WW_ecm240
$Z(b\bar{b})H(WW^*)$	Whizard	0.00499	1,200,000	uzp6_ee_ccH_WW_ecm240
$Z(c\bar{c})H(WW^*)$	Whizard	0.00640	1,200,000	uzp6_ee_ssH_WW_ecm240
Diboson backgrounds				
$e^+e^- \rightarrow WW$	Pythia8	16.44	3.7×10^8	p8_ee_WW_ecm240
$e^+e^- \rightarrow ZZ$	Pythia8	1.36	2.1×10^8	p8_ee_ZZ_ecm240
Two-fermion backgrounds				
$e^+e^- \rightarrow q\bar{q} (u, d, c, s, b)$	Whizard	~ 55	$\sim 5 \times 10^8$	wz3p6_ee_uu_ecm240, wz3p6_ee_dd_ecm240, wz3p6_ee_cc_ecm240, wz3p6_ee_ss_ecm240, wz3p6_ee_bb_ecm240
$e^+e^- \rightarrow \tau^+\tau^-$	Whizard	4.67	2.4×10^8	wz3p6_ee_tau_tau_ecm240
Irreducible ZH ($H \rightarrow b\bar{b}, \tau\tau, gg, c\bar{c}, ZZ^*$)				
$Z(q\bar{q})H(\text{other})$	Whizard	0.04171	2,500,000	uzp6_ee_qqH_bbb_ecm240, uzp6_ee_qqH_tttau_ecm240, uzp6_ee_qqH_bgg_ecm240, uzp6_ee_qqH_bcc_ecm240, uzp6_ee_qqH_bZZ_ecm240
$Z(b\bar{b})H(\text{other})$	Whizard	0.02324	2,100,000	uzp6_ee_bbH_bbb_ecm240, uzp6_ee_bbH_tttau_ecm240, uzp6_ee_bbH_bgg_ecm240, uzp6_ee_bbH_bcc_ecm240, uzp6_ee_bbH_bZZ_ecm240
$Z(c\bar{c})H(\text{other})$	Whizard	0.01825	2,600,000	uzp6_ee_ccH_bbb_ecm240, uzp6_ee_ccH_tttau_ecm240, uzp6_ee_ccH_bgg_ecm240, uzp6_ee_ccH_bcc_ecm240, uzp6_ee_ccH_bZZ_ecm240
$Z(s\bar{s})H(\text{other})$	Whizard	0.02343	1,900,000	uzp6_ee_ssH_bbb_ecm240, uzp6_ee_ssH_tttau_ecm240, uzp6_ee_ssH_bgg_ecm240, uzp6_ee_ssH_bcc_ecm240, uzp6_ee_ssH_bZZ_ecm240

Table 2: Event selection criteria.

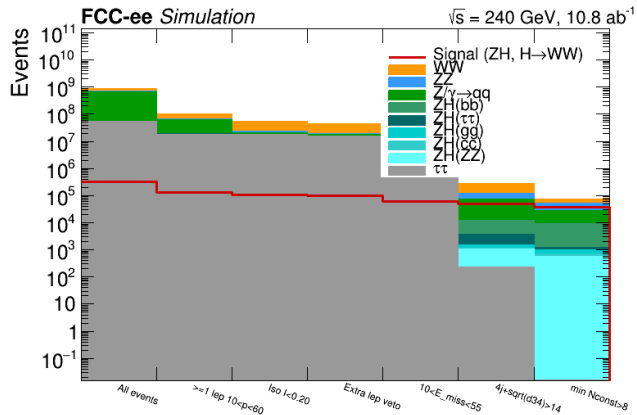
Cut	Requirement	Targeted background
1	≥ 1 lepton with $10 < p_\ell < 60$ GeV	Multi-jet, lepton tails
2	Selected lepton isolation $I_\ell < 0.20$	Non-prompt leptons
3	Veto additional leptons with $p > 20$ GeV	$ZZ \rightarrow \ell\ell qq$, multi-lepton
4	$10 < E_{\text{miss}} < 55$ GeV	Fully hadronic and high-MET tails
5	Exactly 4 jets and $\sqrt{d_{34}} > 14$ GeV	Low-multiplicity / continuum
6	$\min(N_{\text{const}}^{\text{jet}}) > 8$	Sparse fake jets

Table 3: Cutflow: expected yields and cumulative efficiencies [%] at $\mathcal{L} = 10.8 \text{ ab}^{-1}$.

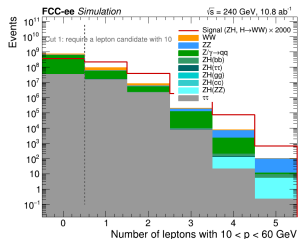
Cut	Signal		WW		ZZ		$q\bar{q}$		$\tau\tau$		ZH(other)	
	Yield	[%]	Yield	[%]	Yield	[%]	Yield	[%]	Yield	[%]	Yield	[%]
All	$3.1 \cdot 10^5$	100	$1.8 \cdot 10^8$	100	$1.5 \cdot 10^7$	100	$5.9 \cdot 10^8$	100	$5 \cdot 10^7$	100	$1.2 \cdot 10^6$	100
1 lep	$1.3 \cdot 10^5$	41.3	$3.5 \cdot 10^7$	19.9	$3.9 \cdot 10^6$	26.8	$4.1 \cdot 10^7$	6.86	$1.7 \cdot 10^7$	34.7	$3 \cdot 10^5$	25.7
Iso	$1.1 \cdot 10^5$	33.8	$3 \cdot 10^7$	16.7	$2.7 \cdot 10^6$	18.6	$2.3 \cdot 10^6$	0.390	$1.7 \cdot 10^7$	33.6	$5.8 \cdot 10^4$	5.04
Veto	$9.8 \cdot 10^4$	31.3	$2.4 \cdot 10^7$	13.8	$7 \cdot 10^5$	4.80	$2.3 \cdot 10^6$	0.389	$1.5 \cdot 10^7$	30.0	$5.3 \cdot 10^4$	4.60
MET	$6.2 \cdot 10^4$	19.6	$9.9 \cdot 10^5$	0.556	$1.6 \cdot 10^5$	1.10	$3.7 \cdot 10^5$	0.062	$5.3 \cdot 10^5$	1.05	$2 \cdot 10^4$	1.76
4 jets	$4.8 \cdot 10^4$	15.4	$1.5 \cdot 10^5$	0.082	$4.7 \cdot 10^4$	0.319	$5.6 \cdot 10^4$	0.0095	217	0.0004	$1.1 \cdot 10^4$	0.990
Nconst	$3.8 \cdot 10^4$	12.2	$2.2 \cdot 10^4$	0.012	$2.2 \cdot 10^4$	0.148	$1.8 \cdot 10^4$	0.0031	0	0	$8.6 \cdot 10^3$	0.750

Nominal baseline ends at Cut6. $m_{Z, \text{cand}}$, $p_{Z, \text{cand}}$, $m_{H, \text{cand}}$ and recoil mass are BDT inputs, not hard filters.

Overall cutflow



Cut 1: at least one lepton with $10 < p < 60$ GeV

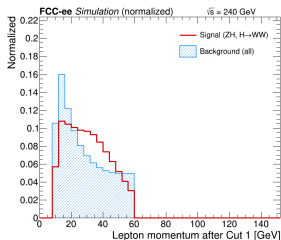


cut1: cut strength

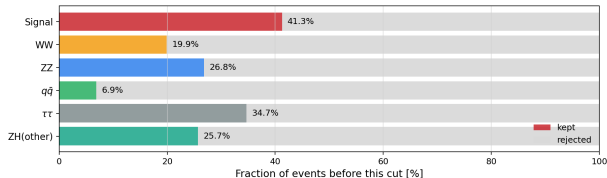
Before: cut0 (All) → After: ≥ 1 lepton, $10 < p_l < 60$ GeV

Counting precision proxy $\sqrt{S+B}/S$: 9.20% → 7.62%; improved by +17.2% (x1.21)

Process	Before	After	Removed	Kept	Rejected	Precision gain
Signal	314,723	129,895	184,828	41.3%	58.7%	+17.2%
WW	1.775e+08	3.536e+07	1.422e+08	19.9%	80.1%	+17.2%
ZZ	1.468e+07	3,940,124	1.074e+07	26.8%	73.2%	+17.2%
q $\bar{q}$	5.941e+08	4.073e+07	5.534e+08	6.9%	93.1%	+17.2%
$\tau\tau$	5.042e+07	1.749e+07	3.293e+07	34.7%	65.3%	+17.2%
ZH(other)	1,151,704	296,231	855,474	25.7%	74.3%	+17.2%

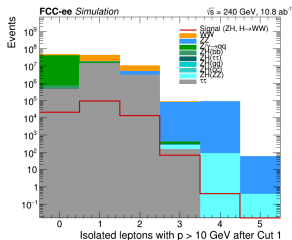
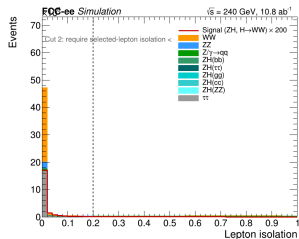


Instantaneous cut effect



Yields are normalized to 10.8 ab^{-1} using the saved cutFlow histograms.

Cut 2: lepton isolation $I_l < 0.20$

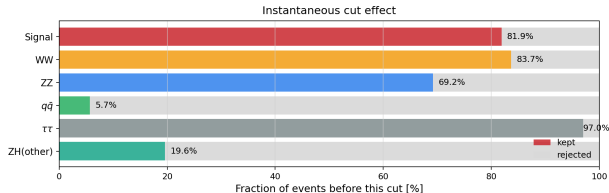


cut2: cut strength

Before: cut1 (≥ 1 lepton, $10 < p_l < 60 \text{ GeV}$) $\rightarrow$ After: Selected lepton isolation $I_l < 0.20$

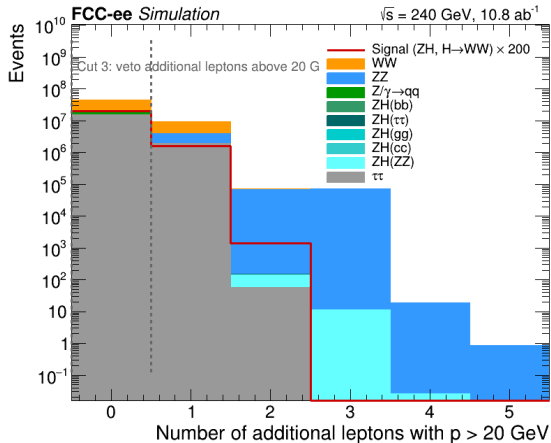
Counting precision proxy $\sqrt{S+B}/S$: 7.62% $\rightarrow$ 6.76%; improved by +11.2% ($\times 1.13$)

Process	Before	After	Removed	Kept	Rejected	Precision gain
Signal	129,895	106,384	23,511	81.9%	18.1%	+11.2%
WW	3.536e+07	2.959e+07	5,770,790	83.7%	16.3%	+11.2%
ZZ	3,940,124	2,727,421	1,212,703	69.2%	30.8%	+11.2%
q $\bar{q}$	4.073e+07	2,316,084	3.841e+07	5.7%	94.3%	+11.2%
$\tau\tau$	1.749e+07	1.696e+07	527,420	97.0%	3.0%	+11.2%
ZH(other)	296,231	58,017	238,214	19.6%	80.4%	+11.2%



Yields are normalized to 10.8 ab^{-1} using the saved cutFlow histograms.

Cut 3: extra isolated lepton veto ($p > 20$ GeV)

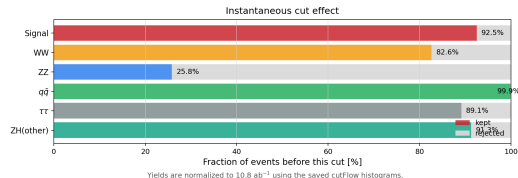


cut3: cut strength

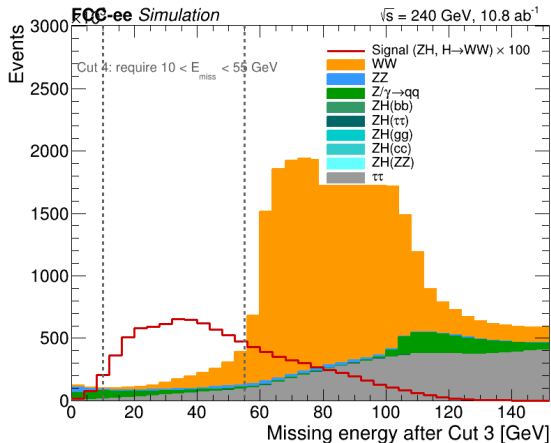
Before: cut2 (Selected lepton isolation $I_l < 0.20$) $\rightarrow$ After: Veto extra lepton, $p > 20$ GeV

Counting precision proxy $\sqrt{S+B}/S$: 6.76% $\rightarrow$ 6.64%; improved by +1.8% (x1.02)

Process	Before	After	Removed	Kept	Rejected	Precision gain
Signal	106,384	98,406	7,978	92.5%	7.5%	+1.8%
WW	2.959e+07	2.444e+07	5,142,109	82.6%	17.4%	+1.8%
ZZ	2,727,421	704,601	2,022,820	25.8%	74.2%	+1.8%
$q\bar{q}$	2,316,084	2,313,401	2,683	99.9%	0.12%	+1.8%
$\tau\tau$	1.696e+07	1.512e+07	1,844,634	89.1%	10.9%	+1.8%
ZH(other)	58,017	52,962	5,055	91.3%	8.7%	+1.8%



Cut 4: missing energy window: $10 < E_{\text{miss}} < 55 \text{ GeV}$

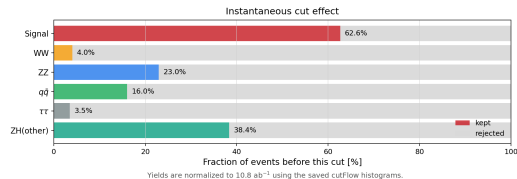


cut4: cut strength

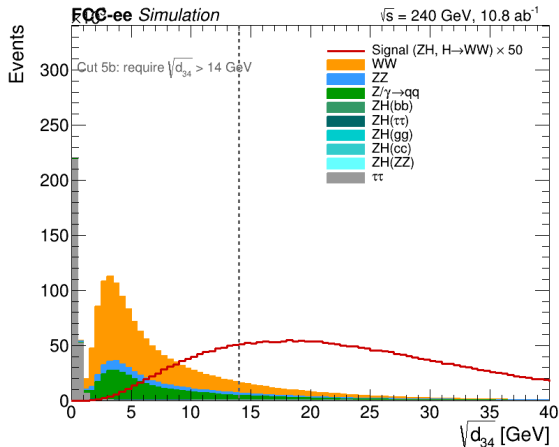
Before: cut3 (Veto extra lepton, $p > 20 \text{ GeV}$) $\rightarrow$ After: $10 < E_{\text{miss}} < 55 \text{ GeV}$

Counting precision proxy $\sqrt{5 + B/S}$: 6.64% $\rightarrow$ 2.37%; improved by +64.3% ($\times 2.80$)

Process	Before	After	Removed	Kept	Rejected	Precision gain
Signal	98,406	61,634	36,772	62.6%	37.4%	+64.3%
WW	2.444e+07	987,530	2.346e+07	4.0%	96.0%	+64.3%
ZZ	704,601	161,893	542,708	23.0%	77.0%	+64.3%
$q\bar{q}$	2,313,401	371,048	1,942,353	16.0%	84.0%	+64.3%
$\tau\tau$	1.512e+07	529,600	1.459e+07	3.5%	96.5%	+64.3%
ZH(other)	52,962	20,315	32,648	38.4%	61.6%	+64.3%



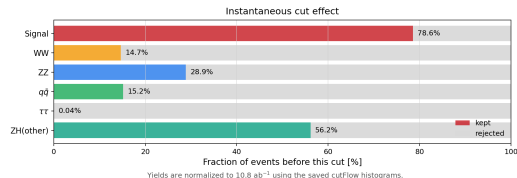
Cut 5: $N = 4$ Durham jets and $\sqrt{d_{34}} > 14$ GeV



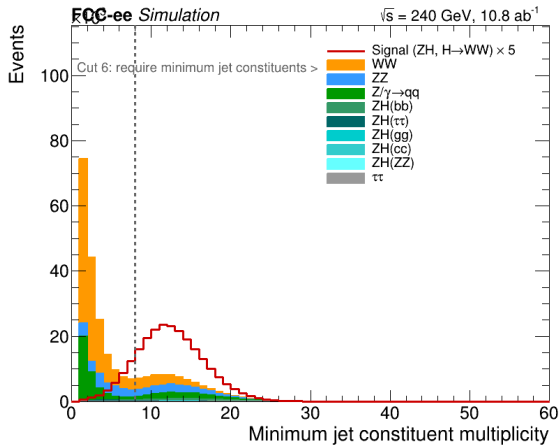
cut5: cut strength

Before: cut4 ($10 < E_{\text{miss}} < 55$ GeV) $\rightarrow$ After: Exclusive $N = 4$ jets + $\sqrt{d_{34}} > 14$ GeV
 Counting precision proxy $\sqrt{S} + B/S$: 2.37% $\rightarrow$ 1.15%; improved by +51.6% ($\times 2.07$)

Process	Before	After	Removed	Kept	Rejected	Precision gain
Signal	61,634	48,421	13,213	78.6%	21.4%	+51.6%
WW	987,530	145,064	842,466	14.7%	85.3%	+51.6%
ZZ	161,893	46,769	115,124	28.9%	71.1%	+51.6%
qq	371,048	56,299	314,748	15.2%	84.8%	+51.6%
$\tau\tau$	529,600	216.8	529,383	0.04%	100.0%	+51.6%
ZH(other)	20,315	11,407	8,907	56.2%	43.8%	+51.6%



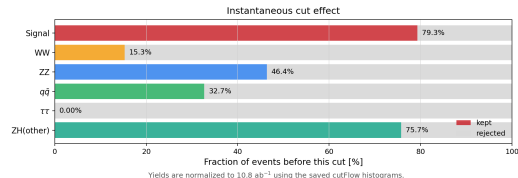
Cut 6: $\min(N_{\text{const}}^{\text{jet}}) > 8$



cut6: cut strength

Before: cut5 (Exclusive $N = 4$ jets + $\sqrt{d_{34}} > 14 \text{ GeV}$) $\rightarrow$ After: $\min(N_{\text{const}}^{\text{jet}}) > 8$
 Counting precision proxy $\sqrt{S+B}/S$: 1.15% $\rightarrow$ 0.86%; improved by +24.9% (x1.33)

Process	Before	After	Removed	Kept	Rejected	Precision gain
Signal	48,421	38,409	10,013	79.3%	20.7%	+24.9%
WW	145,064	22,133	122,931	15.3%	84.7%	+24.9%
ZZ	46,769	21,703	25,066	46.4%	53.6%	+24.9%
q $\bar{q}$	56,299	18,405	37,894	32.7%	67.3%	+24.9%
$\tau\tau$	216.8	0	216.8	0.00%	100.0%	+24.9%
ZH(other)	11,407	8,639	2,769	75.7%	24.3%	+24.9%



Z-Priority Jet Pairing

Challenge

4 jets must be paired into Z on-shell and W^* off-shell candidates; there are 3 possible disjoint pairings for 4 jets.

Algorithm: “Z-priority” pairing

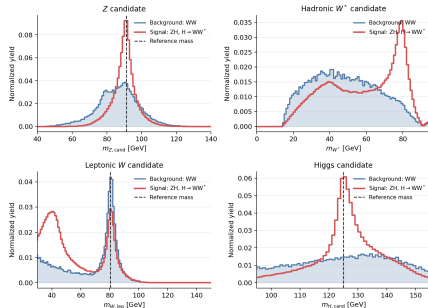
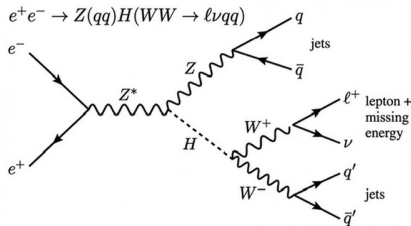
- Among 3 pairings, choose the pair with m_{jj} closest to $m_Z = 91.2$ GeV.
- Assign this pair as the Z candidate.
- Remaining 2 jets form the hadronic W^* candidate.
- Key advantage: does **not** constrain the off-shell W^* mass.

Why not constrain W^* ?

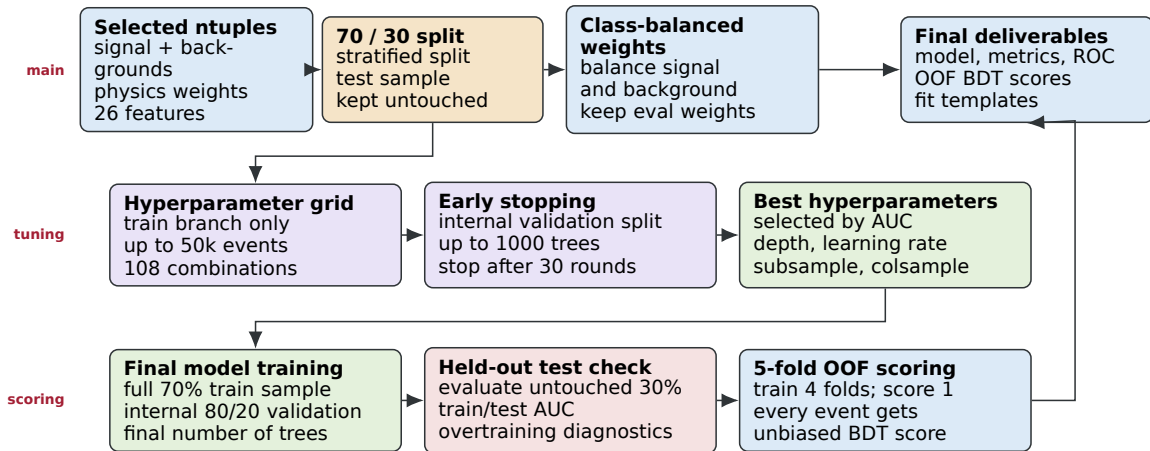
- W^* is far off-shell, peaking near 38 GeV, not $m_W = 80$ GeV.
- Forcing $m_{jj} \sim m_W$ would bias the pairing and lose events.
- Z is on-shell, giving a clean anchor.

Validation after pairing

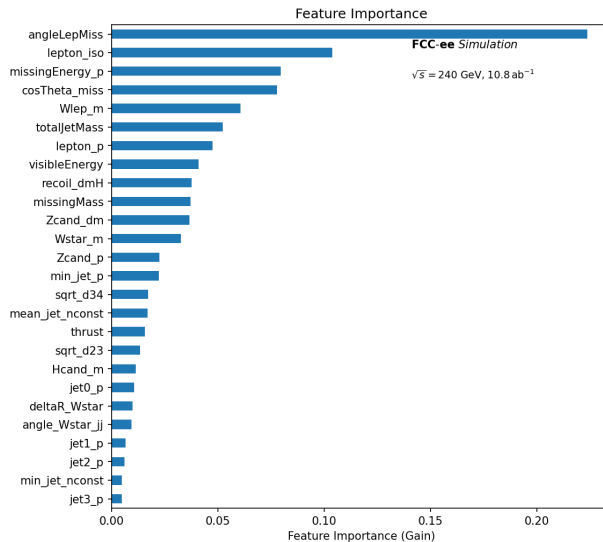
Z candidate peaks at 90.5 GeV with 4.3 GeV core width; leptonic W peaks at m_W ; Higgs candidate peaks at 125.5 GeV. The hadronic W^* spectrum stays broad, with an on-shell component when the leptonic W is the off-shell one.

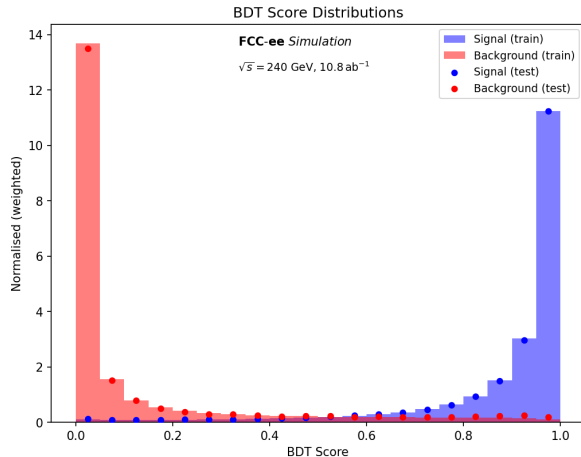
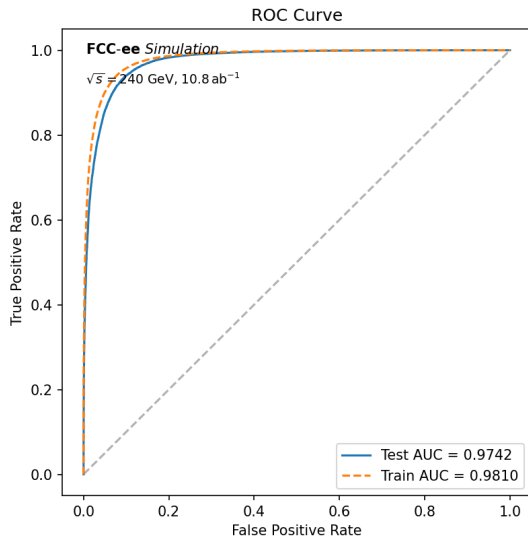


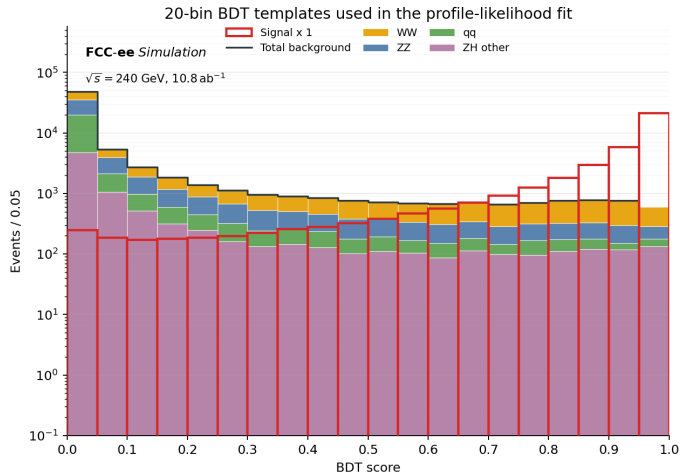
BDT sample after selection: Signal = 38408.6 Background = 70879.6



Feature rank: 26 features







Observed / expected BDT score histogram:

$$N_{\text{obs}}(s) = \mu S(s) + N_{\text{WW}}(s) + N_{\text{ZZ}}(s) + N_{\text{q}\bar{\text{q}}}(s) + N_{\text{ZH,other}}(s)$$

$$\mu = 1 \pm 0.00595$$

$$\frac{\delta\mu}{\mu} = \frac{0.00595}{1} = \mathbf{0.595\%}$$

1% normalisation nuisance per background group + shared MC-template statistical nuisances; MINUIT/HESSE errors validated by a profile-likelihood scan.

Without the MC-template statistical nuisances: 0.564% (statistical floor).